

FKBP5 is associated with amygdala volume in the human brain and mood state: A voxel-based morphometry (VBM) study

Hirofumi Hirakawa, Jotaro Akiyoshi, Masaaki Muronaga, Yoshihiro Tanaka, Yoshinobu Ishitobi, Ayako Inoue, Harumi Oshita, Saeko Aizawa, Koji Masuda, Haruka Higuma, Masayuki Kanehisa, Taiga Ninomiya & Yoshihisa Kawano

To cite this article: Hirofumi Hirakawa, Jotaro Akiyoshi, Masaaki Muronaga, Yoshihiro Tanaka, Yoshinobu Ishitobi, Ayako Inoue, Harumi Oshita, Saeko Aizawa, Koji Masuda, Haruka Higuma, Masayuki Kanehisa, Taiga Ninomiya & Yoshihisa Kawano (2016) FKBP5 is associated with amygdala volume in the human brain and mood state: A voxel-based morphometry (VBM) study, *International Journal of Psychiatry in Clinical Practice*, 20:2, 106-115, DOI: [10.3109/13651501.2016.1144772](https://doi.org/10.3109/13651501.2016.1144772)

To link to this article: <https://doi.org/10.3109/13651501.2016.1144772>



Published online: 16 Mar 2016.



Submit your article to this journal [↗](#)



Article views: 375



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 7 View citing articles [↗](#)

ORIGINAL ARTICLE

FKBP5 is associated with amygdala volume in the human brain and mood state: A voxel-based morphometry (VBM) study

Hirofumi Hirakawa^a, Jotaro Akiyoshi^a, Masaaki Muronaga^a, Yoshihiro Tanaka^a, Yoshinobu Ishitobi^a, Ayako Inoue^a, Harumi Oshita^b, Saeko Aizawa^a, Koji Masuda^a, Haruka Higuma^a, Masayuki Kanehisa^a, Taiga Ninomiya^a and Yoshihisa Kawano^c

^aDepartment of Neuropsychiatry, Oita University Faculty of Medicine, Hasama-Machi, Oita, Japan; ^bDepartment of Applied Linguistics, Oita University Faculty of Medicine, Hasama-Machi, Oita, Japan; ^cKawano Neurosurgery Hospital, Mori-Machi, Oita, Japan

ABSTRACT

The present study was to investigate the effects of 6 FK506 binding protein 51 (FKBP5) single nucleotide polymorphisms (SNPs) on brain structure using voxel-based morphometry (VBM) and the psychological tests to psychological stress. We genotyped 112 healthy controls with respect to 6 SNPs (rs) of FKBP5. We examined the Beck Depression Inventory and the State (STAI-S) and Trait (STAI-T) versions of the Spielberger Anxiety Inventory and the Profile of Mood States (POMS) to evaluate mood. The right amygdala was larger in subjects with the minor allele (C) of rs3800373 and rs992105 and the minor allele (T) of rs1360780. The right middle orbitofrontal region in those with the minor allele (C) of rs3800373 and the right inferior orbitofrontal region in those with the minor allele (T) of rs9470080 was larger. Both the amygdala volumes were associated significantly with FKBP5 SNPs. We found significant relationships between factors in POMS and the right and left amygdala and left insula. Our results suggest that FKBP5 SNPs are associated with the alternations of volumes in right amygdala and the right middle and inferior orbitofrontal region. Genetic variants of FKBP5 may be associated with depressive and anxiety state via differential effects on amygdala and orbitofrontal region.

ARTICLE HISTORY

Received 6 May 2015
Revised 30 November 2015
Accepted 15 January 2016
Published online 15 March 2016

KEYWORDS

Amygdala; FKBP5; genetic susceptibility; genetics; magnetic resonance imaging; POMS

Introduction

Stress has been associated with the onset, symptoms and course of psychiatric illness (Read et al. 2005; Khashan et al. 2008; Arseneault et al. 2011). Dysregulation of the hypothalamic–pituitary–adrenal (HPA) axis has been observed in patients with psychiatric disorders, both under basal conditions and during stress (Gallagher et al. 2007; Bradley et al. 2010). Post-mortem analysis has revealed decreased messenger ribonucleic acid (mRNA) expression of the glucocorticoid receptor (GR), the crucial receptor that is responsible for stress-induced cortisol secretion (Sinclair et al. 2012a, b).

We investigated the role of FK506 binding protein 51 (FKBP5) in modulating the reaction of the HPA axis to social stress (Mahon et al., 2013). This protein is a negative regulator of glucocorticoids that reduces their binding affinity for the GR (Denny et al. 2000). When glucocorticoids bind to the GR, the receptor complex is stimulated, and FKBP51 is exchanged for FK506 binding protein 52 (FKBP52 or FKBP4), a positive regulator of GR

signaling (Riggs et al. 2003). We postulated that these associations are at least partly due to the influence of the observed FKBP5 polymorphism and the subsequent changes in the GR-mediated HPA axis.

Stressful life events (SLEs) have been consistently implicated in the pathophysiology of mood and anxiety disorders (Kendler et al. 1999). The stress response is induced by exposure to stressors and results in catecholamine and glucocorticoid secretion, which are the primary effectors of the stress system. GR function is regulated by a multi-protein complex composed of several FKBP5 variants that have been associated with an increased recurrence of depression and a rapid response to antidepressant treatment (Binder et al. 2004; Kirchheiner et al. 2008; Lekman et al. 2008). The T allele of rs1360780 occurs more frequently among patients affected by major depressive disorder (MDD) with comorbid anxiety disorders (Minelli et al. 2013). The T allele is more often associated with personality traits

that lead to increased vulnerability to anxiety. Decreased GR sensitivity in MDD has only been observed in patients with an FKBP5 risk allele, such as the minor T allele of rs1360780 (Vreeburg et al. 2009; Menke et al. 2013).

Basic animal investigations have demonstrated that stress and anxiety during early development or during the infant or juvenile periods can lead to long-lasting changes in brain structure and function. The most prominent of these changes have been detected in the amygdala, an area that is central to the brain's emotional circuitry (LeDoux 2000; Dolan and Vuilleumier 2003; McEwen 2003; Lupien et al. 2009; Pessoa and Adolphs 2010). Stress and anxiety have also been associated with an enlarged amygdala, particularly the basolateral nuclei, resulting from the persistent interaction of stress-sensitive hormones and experience-dependent plasticity in the growing animal brain (Mitra et al. 2005, 2008; Vyas et al. 2006; Lupien et al. 2009). Generalised anxiety disorder has also been reported to be associated with an increased size of the adult amygdala (Etkin et al. 2007; Schienle et al. 2011; Hettema et al. 2012), as well as in healthy adults with high trait anxiety (Baur et al. 2012; Holmes et al. 2012).

The amygdala is the core area for neural emotional processing and plays fundamental roles in mood and anxiety disorders (Drevets et al. 2008). This structure is involved in the regulation of stress responses by the excitatory effects of the HPA axis (Feldman and Weidenfeld 1998). The main effect of FKBP5 occurs within the left amygdala, which has the highest activity, largest volume and increased coupling with the left hippocampus and the orbitofrontal cortex (OFC; Holz et al. 2014).

We focused on brain regions involved in the initial processing of emotional information (amygdala) and investigated the influence of emotional information (amygdala and OFC) on regions related to appraisal and decision making (Wright et al. 2007). The aim of our study was to investigate whether this polymorphism might be associated with altered brain structure and psychological stress in 112 healthy controls.

Material and methods

Subjects

The study included 112 non-clinical Japanese volunteers from an affective and anxiety neuroimaging study. The study population consisted of healthy volunteers with no history of mental illness. None of the participants were taking any medications at the time of the tests. The exclusion criteria included the following: (1) a history of or current axis I pathology as defined by the Diagnostic and Statistical Manual of Mental Disorders, 4th Edition

(DSM IV); (2) any use of psychoactive drugs; (3) any history of or concurrent major internal or neurological disorder; (4) alcohol and/or drug dependency or recent abuse (past year); (5) hypertension; and (6) general contraindications to magnetic resonance imaging (MRI). All subjects were biologically unrelated. All participants were screened using the Japanese version of the Mini International Neuropsychiatric Interview (M.I.N.I.) (Otsubo et al. 2005) and underwent unstructured interviews conducted by a research psychiatrist. The study was carried out in accordance with the Declaration of Helsinki and was approved by the Ethical Review Boards of the Oita University Faculty of Medicine. After providing a description of the study, written informed consent was obtained from each subject.

SNP genotyping

Deoxyribonucleic acid (DNA) was extracted from blood samples using a DNA extractor SP Kit (WAKO, Osaka, Japan). The single nucleotide polymorphisms (SNPs) were genotyped using the TaqMan 50 exonuclease allelic discrimination assay (Applied Biosystems, Foster City, CA). The following reference SNPs (rs) were tested (A, C, G, and T refer to the alleles under the SNP database nomenclature system): rs3800373, A/C; rs992105, A/C; rs9296158, A/G; rs1360780, C/T; rs9470080, C/T; rs2766534, G/T. According to the manufacturer's instructions, 10 µL of a PCR reaction mixture containing 2 ng of genomic DNA was analysed using the TaqMan PCR kit including a universal PCR master mix. Deviations of the genotype distributions from the Hardy-Weinberg equilibrium (HWE) were assessed with the χ^2 -test for goodness of fit.

Magnetic resonance imaging procedure

MRI scans were performed using a Siemens 3.0-T scanner (Verena, Siemens Medical System) in the Magnetic Resonance Center of Kawano Neurosurgical Hospital. The subjects were asked to lie in the scanner with their eyes closed. A standard birdcage head coil was used with restraining foam pads placed on either side of the head to minimise head motion, and cotton ear plugs were used to reduce noise. For the morphometric study, high spatial resolution, three-dimensional (3D) T1-weighted brain images were obtained by scanning in the sagittal plane (echo time [TE]/repetition time [TR], slice thickness = 1 mm, gap = 0 mm, repetition time (TR) = 2300 ms, echo time (TE) = 2.03 ms, field of view (FOV) = 256 × 256 cm, flip angle = 9, matrix size = 256 × 256, and slices = 176). Raw 3D T1-weighted volume data were transferred to an appropriate workstation.

A preprocessing voxel-based morphometry (VBM) step from a statistical parametric mapping (SPM) protocol was improved using the DARTEL registration method (Ashburner 2007). This technique was more deformable, and it notably improved the small inner structure realignment (Yassa and Stark 2009). Segmentation, normalisation and modulation were performed using DARTEL as a toolbox for SPM8 (Wellcome Department of Imaging Neuroscience, London, UK) running on MATLAB R2013a software (MathWorks, Natick, MA). Gray matter probability values were smoothed using an 8-mm full-width, half-maximum Gaussian kernel.

Psychological tests

We administered the Beck Depression Inventory and the State-Trait Anxiety Inventory (STAI-S and STAI-T) versions of the Spielberger Anxiety Inventory to all participants to evaluate state and trait anxiety (Beck et al. 1979; Spielberger 1983). The STAI has demonstrated well to excellent reliability in trait- and state-anxiety (Spielberger et al. 1970). We also administered the Profile of Mood States to evaluate mood (POMS; McNair et al. 1992). The POMS was designed for use with non-depressed populations and is perceptive to subclinical variations in mood states.

Statistical analyses

Data were presented as mean $\pm$ standard deviation and analysed using the SPSS statistical package version 22 (SPSS Inc., Chicago, IL, USA). A one-way analysis of variance was used for the analysis of age, education years, BDI, STAI, and POMS among the six genotype groups. The χ^2 -test was used for the analysis of gender distribution and the relationship between FKBP5 polymorphism and VBM. Differences in the gray matter volume among the genotype groups were determined using analysis of covariance (ANCOVA), controlling for age, gender, years of education, and whole-brain volume. A seed level of <0.001 (uncorrected) and a cluster size over 759, which was calculated by the AlphaSim programme in AFNI (<http://afni.nimh.nih.gov/afni/>) for cluster-based multiple comparison correction, were selected as criteria. Pearson correlations were used to determine the associations between brain structures and the POMS. Logistic regression analysis was used to identify possible demographic variables and independent predictors of FKBP5 SNPs.

Results

There were no significant differences in the mean age, years of education, or gender distribution among the six

genotype groups. The obtained genotype distributions (rs3800373, rs992105, rs9296158, rs1360780, rs9470080, and rs2766534) did not deviate significantly from the HWE.

Differences in the grey matter volumes among the genotyped groups

SNPs rs3800373, rs992105, rs1360780, and rs9470080: There were significant differences in the right amygdala, $F(1, 2) = 4.14$, $p = 0.02$, and the right-middle and inferior orbitofrontal cortices, $F(1, 2) = 3.72$, $p = 0.03$; $F(1, 2) = 3.73$, $p = 0.03$, associated with the rs3800373 SNP (Figures 1 and 2). Figure 1 shows an MRI demonstrating the higher grey matter volume (GMV) in the right amygdala associated with the C allele (rs3800373; CC carriers $>$ AC and AA). Figure 2 shows an MRI depicting the larger GMV in the right-middle and inferior orbitofrontal cortices associated with the C allele (rs3800373; CC carriers $>$ AC and AA).

There was a significant difference in the right amygdala, $F(1, 2) = 3.50$, $p = 0.03$, associated with the rs992105 SNP (Figure 3). Figure 3 shows an MRI exhibiting a larger GMV in the right-middle and inferior OFC associated with the C allele (rs992105; CC carriers $>$ AC and AA).

There was a significant difference in the right amygdala for the rs1360780 SNP, $F(1, 2) = 5.21$, $p = 0.001$ (Figure 4). Figure 4 shows an MRI depicting a larger GMV in the right amygdala associated with the T allele (rs1360780; TT carriers $>$ CT and CC).

There was a significant difference in the right inferior OFC for the rs9470080 SNP, $F(1, 2) = 5.65$, $p = 0.005$ (Figure 5). Figure 5 shows an MRI revealing a larger GMV in the right inferior OFC associated with the T allele (rs9470080; TT carriers $>$ CT and CC).

SNPs rs929615 and rs2766534: we found no statistically significant alterations of the GMV in several cortical regions associated with the rs9296158 and rs2766534 SNPs (Supplementary Table S1).

Differences in the psychological tests among the genotyped groups

The BDI scores in the AA group of rs992105 were higher than those in the AC and CC groups ($p < 0.049$) (Supplementary Table S2). There were no significant differences among STAI scores in the rs3800373, rs992105, rs9296158, rs1360780, rs9470080, and rs2766534 SNP groups (Supplementary Table S2). There were no significant differences among POMS scores in the rs3800373, rs992105, rs9296158, rs1360780, rs9470080 and rs2766534 SNP groups (Supplementary Table S2).

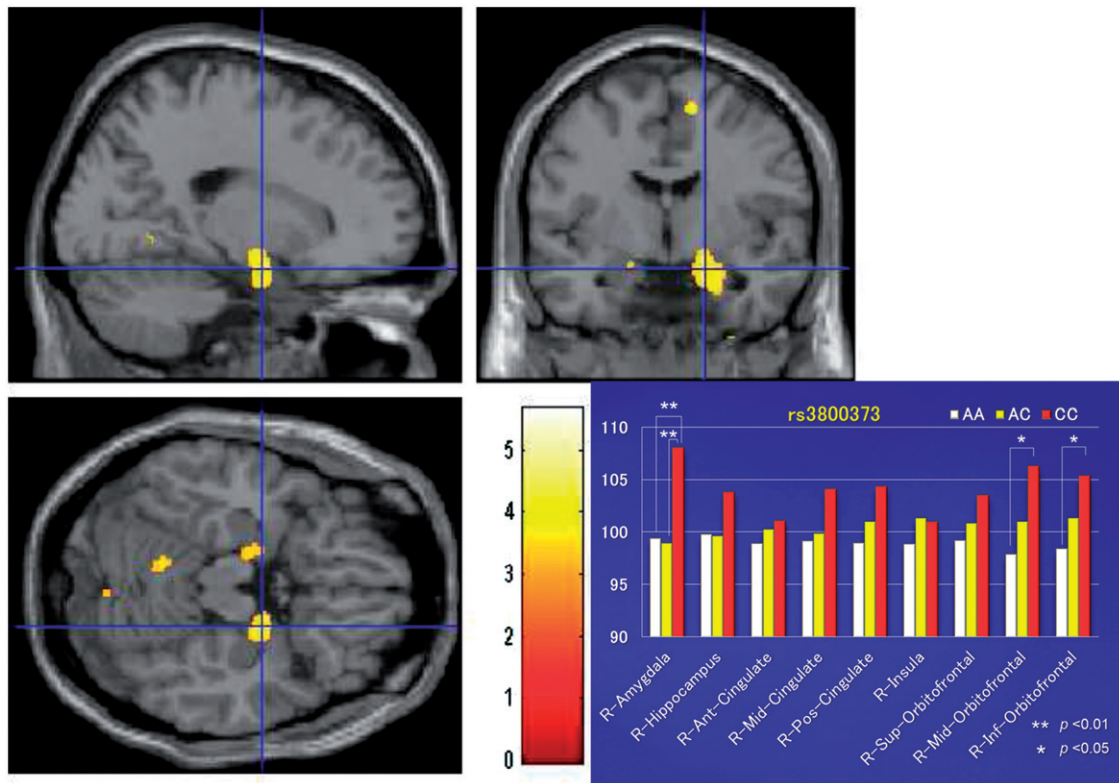


Figure 1. Magnetic resonance imaging (MRI) exhibiting higher grey matter volume in the right amygdala associated with the C allele (rs3800373; CC carriers > AC and AA). For illustrative purposes, the statistical threshold was set at $p < 0.001$ (uncorrected). There were no differences between the CC carriers and the AC + AA carriers with regard to tension–anxiety, depression–dejection, anger–hostility, vigor, fatigue or confusion.

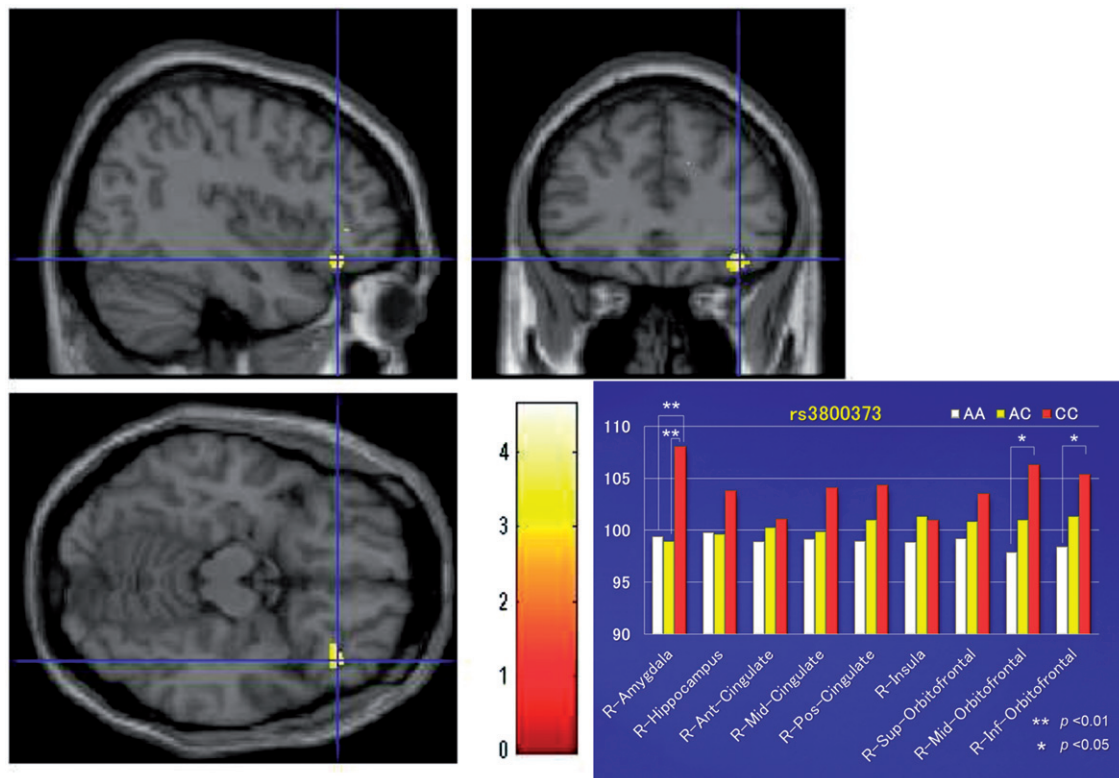


Figure 2. Magnetic resonance imaging (MRI) exhibiting higher grey matter volumes in the right middle and inferior orbitofrontal cortices associated with the C allele (rs3800373; CC carriers > AC and AA). For illustrative purposes, the statistical threshold was set at $p < 0.001$ (uncorrected). There were no differences between the CC carriers and the AC + AA carriers with regard to tension–anxiety, depression–dejection, anger–hostility, vigor, fatigue or confusion.

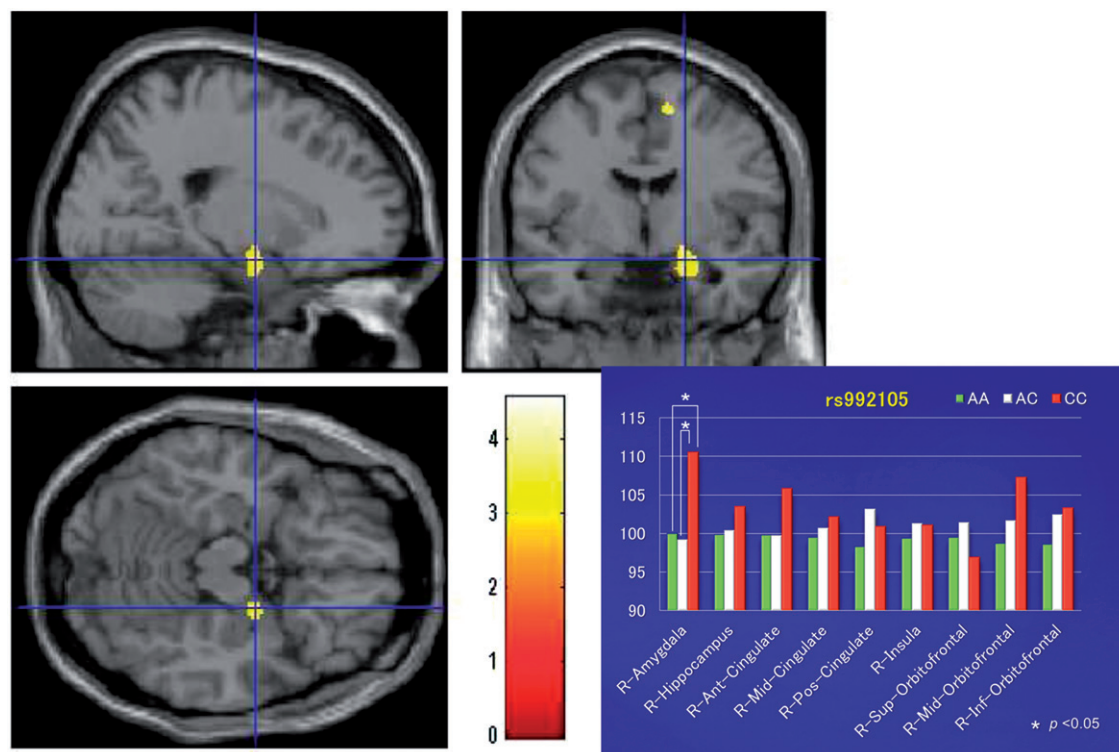


Figure 3. Magnetic resonance imaging (MRI) exhibiting higher grey matter volume in the right amygdala associated with the C allele (rs992105; CC carriers > AC and AA). For illustrative purposes, the statistical threshold was set at $p < 0.001$ (uncorrected). The scores for tension–anxiety and depression–dejection were significantly higher among the CC carriers compared with the AC + AA carriers ($p < 0.01$). There scores for vigor were significantly lower among the CC carriers compared with the AC + AA carriers ($p < 0.05$). There were no differences between the CC carriers and the AC + AA carriers with regard to anger–hostility, fatigue or confusion.

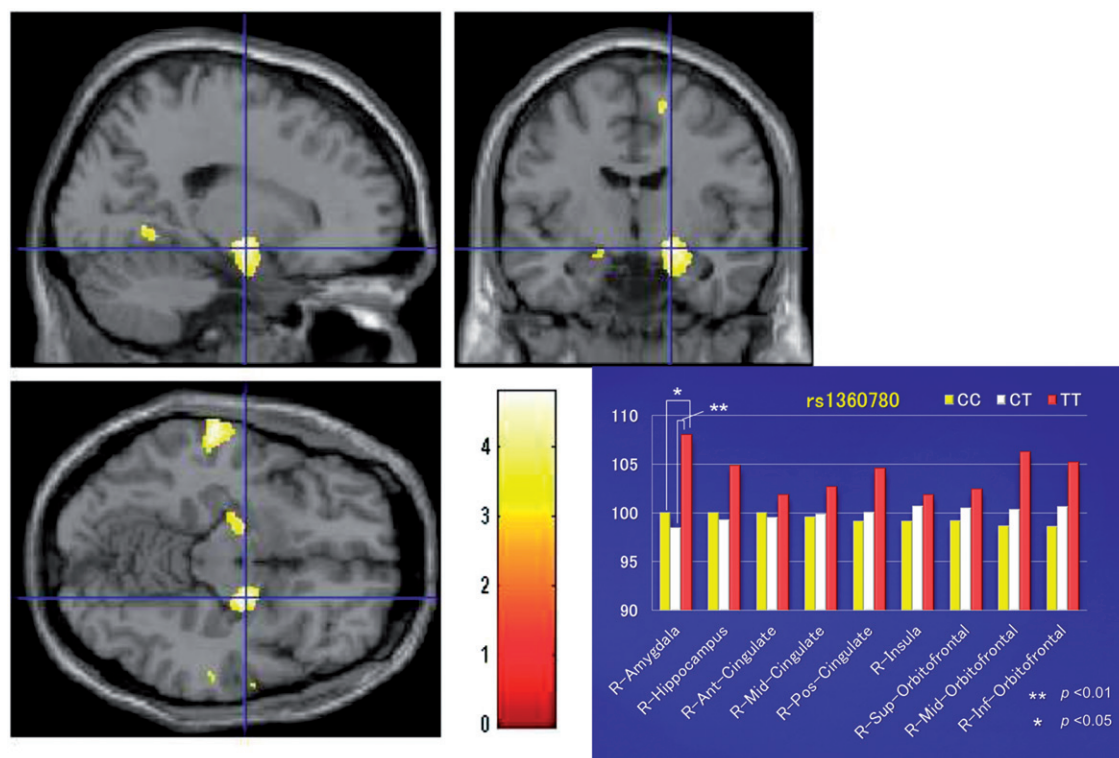


Figure 4. Magnetic resonance imaging (MRI) exhibiting higher grey matter volume in the right amygdala associated with the T allele (rs1360780; TT carriers > CT and CC). For illustrative purposes, the statistical threshold was set at $p < 0.001$ (uncorrected). There were no differences between the TT carriers and the TC + TT carriers with regard to tension–anxiety, depression–dejection, anger–hostility, vigor, fatigue or confusion.

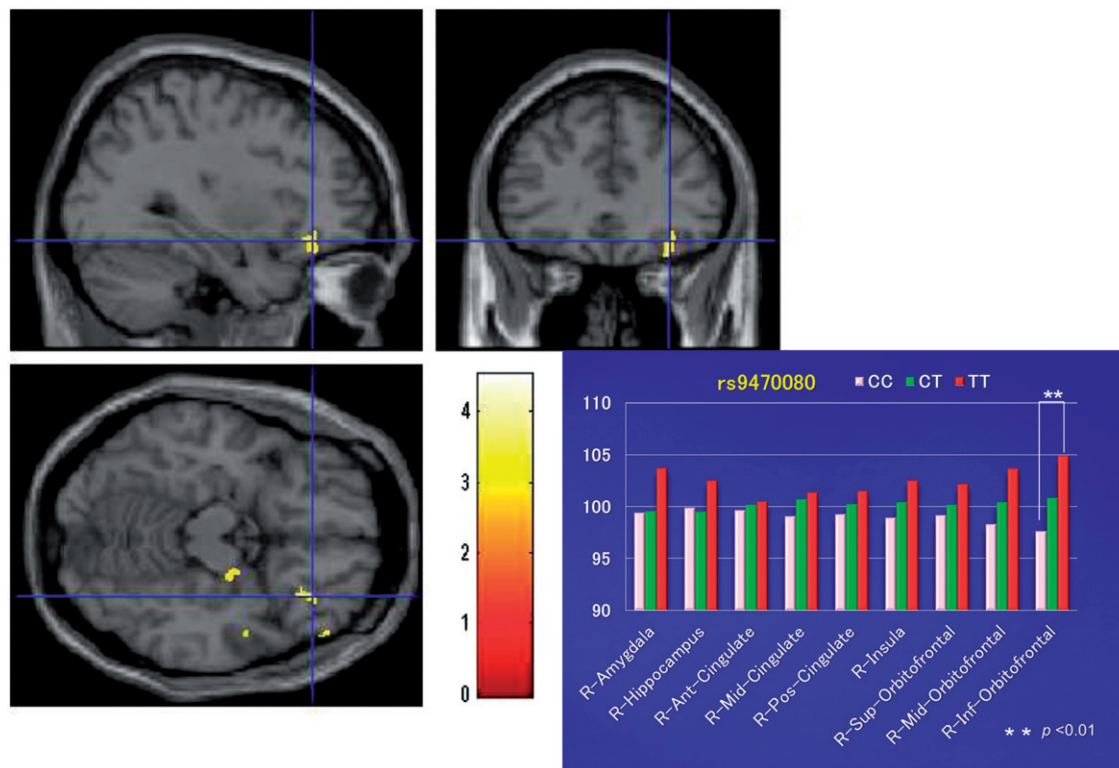


Figure 5. Magnetic resonance imaging (MRI) exhibiting higher grey matter volume in the right inferior orbitofrontal cortex associated with the T allele (rs9470080; TT carriers > CT and CC). For illustrative purposes, the statistical threshold was set at $p < 0.001$ (uncorrected). The scores for anger–hostility, vigor and fatigue were significantly lower among the TT carriers compared with the CT + TT carriers ($p < 0.05$). There were no differences between the TT carriers and the CT + TT carriers with regard to tension–anxiety, depression–dejection, fatigue or confusion.

Table 1. Relationship between STAI and VBM.

	Trait	State
Right		
Amygdala	0.16	0.20*
Hippocampus	0.08	0.05
Anterior cingulate	0.08	0.08
Middle cingulate	0.03	−0.05
Posterior cingulate	0.05	−0.01
Insula	0.14	0.08
Superior orbitofrontal	−0.02	−0.03
Middle orbitofrontal	−0.01	−0.13
Inferior orbitofrontal	0.06	0.06
Left		
Amygdala	0.17*	0.18*
Hippocampus	0.14	0.07
Anterior cingulate	0.03	0.09
Middle cingulate	−0.06	−0.12
Posterior cingulate	−0.05	−0.08
Insula	0.11	0.09
Superior orbitofrontal	0.01	0.00
Middle orbitofrontal	0.09	−0.06
Inferior orbitofrontal	0.12	0.07

STAI, State-Trait Anxiety Inventory; VBM, voxel-based morphometry.

* $p < 0.05$.

Relationship between the POMS and VBM

We found significant relationships between depression–dejection, anger–hostility and fatigue and the right amygdala. There were significant relationships between

tension–anxiety, depression–dejection, anger–hostility and fatigue and the left amygdala. There was a significant relationship between anger–hostility and the left insula (Table 2).

Predictors of FKBP5 SNPs

According to the logistic regression analysis, two variables were identified as independent predictors of FKBP5 SNPs: VBM and the POMS. These variables explained 19.29% of the variation in rs3800373 (AA), 18.61% of the variation in rs3800373 (AC), 14.60% of the variation in rs3800373 (AA), and 12.79% of the variation in rs9470080 (CC). The right and left amygdala volumes were associated significantly with FKBP5 SNPs.

Discussion

We identified variations in gray matter brain volume that were independently associated with 4 SNPs (rs3800373, rs992105, rs1360780 and rs9470080) of FKBP5. Specifically, MRI revealed increases in GMV that were associated with the FKBP5 SNPs.

Table 2. Relationship between POMS and VBM.

	T-A	D	A-H	V	F	C
Right						
Amygdala	0.16	0.20*	0.19*	0.10	0.18**	0.07
Hippocampus	0.08	0.05	0.11	0.04	0.09	0.04
Anterior cingulate	0.08	0.08	0.09	0.12	0.04	0.04
Middle cingulate	0.03	-0.05	0.00	0.02	-0.05	0.04
Posterior cingulate	0.05	-0.01	-0.06	0.03	0.08	0.05
Insula	0.14	0.08	0.14	0.04	0.12	0.07
Superior orbitofrontal	-0.02	-0.03	-0.02	0.02	0.04	-0.07
Middle orbitofrontal	-0.01	-0.13	0.02	0.07	-0.06	-0.15
Inferior orbitofrontal	0.06	0.06	-0.02	0.00	0.01	-0.08
Left						
Amygdala	0.17*	0.18*	0.22	0.09	0.23**	-0.04
Hippocampus	0.14	0.07	0.16	0.07	0.14	0.00
Anterior cingulate	0.03	0.09	0.13	0.10	0.07	0.05
Middle cingulate	-0.06	-0.12	-0.03	0.05	-0.13	-0.03
Posterior cingulate	-0.05	-0.08	-0.06	0.13	-0.01	-0.14
Insula	0.11	0.09	0.19*	0.12	0.09	0.02
Superior orbitofrontal	0.01	0.00	-0.05	-0.03	-0.02	0.02
Middle orbitofrontal	0.09	-0.06	0.08	0.04	0.04	-0.01
Inferior orbitofrontal	0.12	0.07	0.05	0.05	0.04	0.06

POMS, Profile of Mood States; VBM, voxel-based morphometry; T-A, tension-anxiety; D, depression-dejection; A-H, anger-hostility; V, vigor; F, fatigue; C, confusion.

* $p < 0.05$, ** $p < 0.01$.

The right amygdala volume was associated with the C allele (rs3800373; CC carriers > AC and AA). rs3800373 has been reported to interact with emotional neglect and right-dorsal amygdala reactivity (White et al. 2012). The right-middle and inferior orbitofrontal cortices were associated with the C allele (rs3800373; CC carriers > AC and AA). The right amygdala was associated with the C allele (rs992105; CC carriers > AC and AA). There were significantly higher levels of tension-anxiety and depression-dejection among the CC carriers of the rs992105 allele than among the AC + AA carriers. There was also a significantly lower level of vigor among the CC carriers than the AC + AA carriers ($p < 0.05$). This is a novel report of the associations among the amygdala, the rs992105 allele, and psychological effects. The carriers of the CC genotype for the rs992105 SNP exhibited anxiety and depression and enlargement of the right amygdala. Some reports have linked the right amygdala to an early, rapid, and possibly more automatic recognition of emotional stimuli with less adaptation and, eventually, a favored reaction to negative stimuli (Sergerie et al. 2008; Baeken et al. 2010; Dyck et al. 2011). Conversely, the left amygdala was thought to be involved in more complex and intricate stimulus evaluation, as occurs with other cognitive stimuli (Sergerie et al. 2008; Dyck et al. 2011).

The right amygdala was associated with the T allele (rs1360780; TT carriers > CT and CC). rs1360780 was associated with emotional neglect and right dorsal amygdala reactivity (White et al. 2012). However, Holz et al. (2014) reported the opposite results, showing a significant increase in T homozygotes and a significant decrease in C homozygotes. Some reasons for this

inconsistency may include differences in sample conformation and the application of the imaging paradigm being used. The rs1360780 T allele was correlated with higher scores for harm avoidance (HA) and lower scores for cooperativeness (CO) compared with individuals lacking the T allele (Shibuya et al. 2010). The T allele was associated with higher scores for HA among females and lower scores for CO among males. Carriers of the rs1360780 T allele exhibit the following: increased FKBP5 protein levels, reduced basal cortisol levels, impaired HPA negative feedback (dexamethasone) and psychosocial stress. Further investigations are necessary to address this inconsistency. rs1360780, but not rs3800373, was significantly associated with depressive status (Lekman et al. 2008). Homozygous minor allele carriers of the rs1360780 SNP exhibited increased FKBP5 protein levels in healthy controls (Binder et al. 2004). rs1360780 has also been associated with the Trier Social Stress Test (TSST) response in an adult sample (Ising et al. 2008).

The right inferior OFC was associated with the T allele (rs9470080; TT carriers > CT and CC). rs9470080 influenced the suicide risk in individuals exposed to high levels of childhood trauma (Roy et al. 2010). For example, rs9470080 was associated with both the area under the receiver operating characteristic curve (AUC) for cortisol and the likelihood of depressive symptoms (Velders et al. 2011). Exposure to childhood trauma specifically influences behavioural dyscontrol and the lifetime history of aggression and violent behaviour (Bevilacqua et al. 2012). Furthermore, rs9470080 interacted with emotional neglect and right dorsal amygdala reactivity (White et al. 2012). The T allele of rs9470080

was more common in PTSD in the context of adverse environmental exposure (Boscarino et al. 2012). FKBP5, including rs9470080, was associated with exposure to child maltreatment and limbic irritability symptoms (Dackis et al. 2012).

We found that state anxiety in STAI was associated with right and left amygdala higher GMV, and trait anxiety in STAI was associated with left amygdala. We also found depression–dejection, anger–hostility and fatigue in POMS were associated with right amygdala, and tension–anxiety, depression–dejection, anger–hostility and fatigue in POMS were associated with left amygdala. Recently, Mincic (2015) reported that higher negative emotionality (NE)-related traits are linked to higher GMV in left amygdala. He suggests that the behavioural phenotype associated to NE traits is displayed in structural differences with in the cortico-limbic system. This higher GMV in amygdala might be associated with CC carrier in rs3800373 and rs992105 and TT carriers in rs1360780.

Recently, stress has been shown to induce serine protease neuropsin activity in the amygdala, thereby inducing anxiety in mice via an FKBP5-dependent mechanism (Attwood et al. 2011). FKBP5 SNPs might contribute to increased sensitivity of the amygdala/HPA axis response to adult stress. The amygdala controls emotional learning in the brain, and FKBP5 expression can be upregulated by GR activation (Scharf et al. 2011).

This study has several limitations. Although all participants who had no history of mental disorder were considered healthy, they were not evaluable for life trauma and life stress events. Because the sample size was small, it was difficult to detect any genotype-dependent differences in this study. Furthermore, our research was limited by the cross-sectional design. A prospective study is necessary to elucidate the changes in GMV volume in specific brain regions during the aging process; indeed, the effect of FKBP5 on GMV volume may vary with time and age. Thus, the enrollment of subjects with the wide range of ages present in the current study may mask unique effects occurring within specific age groups. These limitations should be resolved in future studies.

Conclusions

We found that the FKBP5 SNP could modulate the regional GMV volume in a healthy population and significant alterations in the volume of the amygdala in association with FKBP5 SNPs. These results suggested that FKBP5 SNPs were associated with the right and left amygdala and the left insula structure and

tension–anxiety, depression–dejection, anger–hostility and fatigue in POMS in 112 healthy controls.

Key points

- We examined 6 FKBP5 polymorphisms in 112 healthy controls.
- The brain structural alterations were analysed using the VBM method.
- The volume of amygdala was associated with rs3800373, rs992105 and rs1360780 genotypes.
- The volume of right middle orbitofrontal region was associated with rs3800373 genotype.
- The right inferior orbitofrontal region was associated with rs9470080 genotype.

Acknowledgements

This research was supported by a Grant-in-Aid for Scientific Research (Number 20591371) from the Japanese Society for the Promotion of Science (JSPS). We are grateful to Mr. Shunsuke Kobayashi and Masaaki Muronaga for her excellent technical assistance.

Disclosure Statement

The authors report no conflicts of interest.

References

- Arseneault L, Cannon M, Fisher HL, Polanczyk G, Moffitt TE, Caspi A, et al. 2011. Childhood trauma and children's emerging psychotic symptoms: a genetically sensitive longitudinal cohort study. *Am J Psychiatry* 168:65–72.
- Ashburner J. 2007. A fast diffeomorphic image registration algorithm. *Neuroimage* 38:95–113.
- Attwood BK, Bourgognon J-M, Patel S, Mucha M, Schiavon E, Skrzypiec AE, et al. 2011. Neuropsin cleaves EphB2 in the amygdala to control anxiety. *Nature* 473:372–375.
- Baeken C, De Raedt R, Van Schuerbeek P, Vanderhasselt MA, De Mey J, Bossuyt A, et al. 2010. Right prefrontal HF-rTMS attenuates right amygdala processing of negatively valenced emotional stimuli in healthy females. *Behav Brain Res* 214:450–455.
- Baur V, Hänggi J, Jäncke L. 2012. Volumetric associations between uncinate fasciculus, amygdala, and trait anxiety. *BMC Neurosci* 13:437–446.
- Beck AT, Rush AJ, Shaw BF, Emery G. 1979. *Cognitive therapy of depression*. New York: Guilford Press.
- Bevilacqua L, Carli V, Sarchiapone M, George DK, Goldman D, Roy A, et al. 2012. Interaction between FKBP5 and childhood trauma and risk of aggressive behavior. *Arch Gen Psychiatry* 69:62–70.
- Binder EB, Salyakina D, Lichtner P, Wochnik GM, Ising M, Putz B, et al. 2004. Polymorphisms in FKBP5 are associated with increased recurrence of depressive episodes and rapid response to antidepressant treatment. *Nat Genet* 36:1319–1325.

- Boscarino JA, Erlich PM, Hoffman SN, Zhang X. 2012. Higher FKBP5, COMT, CHRNA5, and CRHR1 allele burdens are associated with PTSD and interact with trauma exposure: implications for neuropsychiatric research and treatment. *Neuropsychiatr Dis Treat* 8:131–139.
- Bradley AJ, Dinan TG. 2010. A systematic review of hypothalamic-pituitary-adrenal axis function in schizophrenia: implications for mortality. *J Psychopharmacol* 24:91–118.
- Dackis MN, Rogosch FA, Oshri A, Cicchetti D. 2012. The role of limbic system irritability in linking history of childhood maltreatment and psychiatric outcomes in low-income, high-risk women: moderation by FK506 binding protein 5 haplotype. *Dev Psychopathol* 24:1237–1252.
- Denny WB, Valentine DL, Reynolds PD, Smith DF, Scammell JG. 2000. Squirrel monkey immunophilin FKBP51 is a potent inhibitor of glucocorticoid receptor binding. *Endocrinology* 141:4107–4113.
- Dolan RJ, Vuilleumier P. 2003. Amygdala automaticity in emotional processing. *Ann N Y Acad Sci* 985:348–355.
- Drevets WC, Price JL, Furey ML. 2008. Brain structural and functional abnormalities in mood disorders: implications for neurocircuitry models of depression. *Brain Struct Funct* 213:93–118.
- Dyck M, Loughhead J, Kellermann T, Boers F, Gur RC, Mathiak K. 2011. Cognitive versus automatic mechanisms of mood induction differentially activate left and right amygdala. *Neuroimage* 54:2503–2513.
- Etkin A, Wager TD. 2007. Functional neuroimaging of anxiety: a meta-analysis of emotional processing in PTSD, social anxiety disorder, and specific phobia. *Am J Psychiatry* 164:1476–1488.
- Feldman S, Weidenfeld J. 1998. The excitatory effects of the amygdala on hypothalamo-pituitary-adrenocortical responses are mediated by hypothalamic norepinephrine, serotonin, and CRF-41. *Brain Res Bull* 45:389–393.
- Gallagher P, Watson S, Smith MS, Young AH, Ferrier IN. 2007. Plasma cortisol-dehydroepiandrosterone (DHEA) ratios in schizophrenia and bipolar disorder. *Schizophr Res* 90:258–265.
- Hettema JM, Kettenmann B, Ahluwalia V, McCarthy C, Kates WR, Schmitt JE, et al. 2012. Pilot multimodal twin imaging study of generalized anxiety disorder. *Depress Anxiety* 29:202–209.
- Holmes AJ, Lee PH, Hollinshead MO, Bakst L, Roffman JL, Smoller JW, et al. 2012. Individual differences in amygdala-medial prefrontal anatomy link negative affect, impaired social functioning, and polygenic depression risk. *J Neurosci* 32:18087–18100.
- Holz NE, Buchmann AF, Boecker R, Blomeyer D, Baumeister S, Wolf I, et al. 2014. Role of FKBP5 in emotion processing: results on amygdala activity, connectivity and volume. *Brain Struct Funct* 220:1355–1368.
- Ising M, Depping AM, Siebertz A, Lucae S, PG U, Kloiber S. 2008. Polymorphisms in the FKBP5 gene region modulate recovery from psychosocial stress in healthy controls. *Eur J Neurosci* 28:389–398.
- Kendler KS, Karkowski LM, Prescott CA. 1999. Causal relationship between stressful life events and the onset of major depression. *Am J Psychiatry* 156:837–841.
- Khashan AS, Abel KM, McNamee R, Pedersen MG, Webb RT, Baker PN, et al. 2008. Higher risk of offspring schizophrenia following antenatal maternal exposure to severe adverse life events. *Arch Gen Psychiatry* 65:146–152.
- Kirchheiner J, Lorch R, Lebedeva E, Seeringer A, Roots I, Sasse J, et al. 2008. Genetic variants in FKBP5 affecting response to antidepressant drug treatment. *Pharmacogenomics* 9:841–846.
- LeDoux JE. 2000. Emotion circuits in the brain. *Annu Rev Neurosci* 23:155–184.
- Lekman M, Laje G, Charney D, Rush AJ, Wilson AF, Sorant AJ, et al. 2008. The FKBP5-gene in depression and treatment response – an association study in the Sequenced Treatment Alternatives to Relieve Depression (STAR*D) Cohort. *Biol Psychiatry* 63:1103–1110.
- Lupien SJ, McEwen BS, Gunnar MR, Heim C. 2009. Effects of stress throughout the lifespan on the brain, behaviour and cognition. *Nat Rev Neurosci* 10:434–445.
- Mahon PB, Zandi PP, Potash JB, Nestadt G, Wand GS. 2013. Genetic association of FKBP5 and CRHR1 with cortisol response to acute psychosocial stress in healthy adults. *Psychopharmacology (Berl)*. 227:231–241.
- McEwen BS. 2003. Early life influences on life-long patterns of behavior and health. *Ment Retard Dev Disabil Res Rev* 9:149–154.
- McNair DM, Lorr M, Droppleman LF. 1992. Revised manual for the profile of mood states. San Diego (CA): Educational and Industrial Testing Services.
- Mincic AM. 2015. Neuroanatomical correlates of negative emotionality-related traits: a systematic review and meta-analysis. *Neuropsychologia* 77:97–118.
- Mitra R, Jadhav S, McEwen BS, Vyas A, Chattarji S. 2005. Stress duration modulates the spatiotemporal patterns of spine formation in the basolateral amygdala. *Proc Natl Acad Sci USA* 102:9371–9376.
- Mitra R, Sapolsky RM. 2008. Acute corticosterone treatment is sufficient to induce anxiety and amygdaloid dendritic hypertrophy. *Proc Natl Acad Sci USA* 105:5573–5578.
- Otsubo T, Tanaka K, Koda R, Shinoda J, Sano N, Tanaka S, et al. 2005. Reliability and validity of Japanese version of the Mini-International Neuropsychiatric Interview. *Psychiatry Clin Neurosci* 59:517–526.
- Pessoa L, Adolphs R. 2010. Emotion processing and the amygdala: from a “low road” to “many roads” of evaluating biological significance. *Nat Rev Neurosci* 11:773–783.
- Read J, van Os J, Morrison AP, Ross CA. 2005. Childhood trauma, psychosis and schizophrenia: a literature review with theoretical and clinical implications. *Acta Psychiatr Scand* 112:330–350.
- Riggs DL, Roberts PJ, Chirillo SC, Cheung-Flynn J, Prapapanich V, Ratajczak T, et al. 2003. The Hsp90-binding peptidylprolyl isomerase FKBP52 potentiates glucocorticoid signaling in vivo. *EMBO J* 22:1158–1167.
- Roy A, Gorodetsky E, Yuan Q, Goldman D, Enoch MA. 2010. Interaction of FKBP5, a stress-related gene, with childhood trauma increases the risk for attempting suicide. *Neuropsychopharmacology* 35:1674–1683.
- Scharf SH, Liebl C, Binder EB, Schmidt MV, Muller MB. 2011. Expression and regulation of the Fkbp5 gene in the adult mouse brain. *PLoS One* 6:e16883.
- Schienze A, Ebner F, Schafer A. 2011. Localized gray matter volume abnormalities in generalized anxiety disorder. *Eur Arch Psychiatry Clin Neurosci* 261:303–307.
- Sergerie K, Chochol C, Armony JL. 2008. The role of the amygdala in emotional processing: a quantitative meta-analysis of functional neuroimaging studies. *Neurosci Biobehav Rev* 32:811–830.

- Shibuya N, Suzuki A, Sadahiro R, Kamata M, Matsumoto Y, Goto K, et al. 2010. Association study between a functional polymorphism of FK506-binding protein 51 (FKBP5) gene and personality traits in healthy subjects. *Neurosci Lett* 485:194–197.
- Sinclair D, Fullerton JM, Webster MJ, Shannon Weickert C. 2012. Glucocorticoid receptor 1B and 1C mRNA transcript alterations in schizophrenia and bipolar disorder, and their possible regulation by GR gene variants. *PLoS One* 7:e31720.
- Sinclair D, Webster MJ, Fullerton JM, Weickert CS. 2012. Glucocorticoid receptor mRNA and protein isoform alterations in the orbitofrontal cortex in schizophrenia and bipolar disorder. *BMC Psychiatry* 12:84.
- Spielberger C. 1983. State-Trait Anxiety Inventory (Form Y). Palo Alto (CA): Mind Garden.
- Spielberger CD, Gorsuch RL, Lushene RE. 1970. Handbook of STAI. Palo Alto, CA: Consulting Psychologist Press.
- Velders FP, Kuningas M, Kumari M, Dekker MJ, Uitterlinden AG, Kirschbaum C, et al. 2011. Genetics of cortisol secretion and depressive symptoms: a candidate gene and genome wide association approach. *Psychoneuroendocrinology* 36: 1053–1061.
- Vreeburg SA, Hoogendijk WJG, van Pelt J, Derijk RH, Verhagen JCM, van Dyck R, et al. 2009. Major depressive disorder and hypothalamic-pituitary-adrenal axis activity: results from a large cohort study. *Arch Gen Psychiatry* 66:617–626.
- Vyas A, Jadhav S, Chattarji S. 2006. Prolonged behavioral stress enhances synaptic connectivity in the basolateral amygdala. *Neuroscience* 143:387–393.
- White MG, Bogdan R, Fisher PM, Munoz KE, Williamson DE, Hariri AR. 2012. FKBP5 and emotional neglect interact to predict individual differences in amygdala reactivity. *Genes Brain Behav* 11:869–878.
- Wright CI, Feczko E, Dickerson B, Williams D. 2007. Neuroanatomical correlates of personality in the elderly. *Neuroimage* 35:263–272.
- Yassa MA, Stark CEL. 2009. A quantitative evaluation of cross-participant registration techniques for MRI studies of the medial temporal lobe. *Neuroimage* 44:319–327.